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A taxonomy of human driver errors and unsafe adaptations in mixed traffic

Olga Mironenko[1*], Hadi Banaee[1] and Amy Loutfi[1]

Abstract
Human driver errors remain a central road-safety concern, but existing error and violation taxonomies were developed mainly for conventional traffic. In mixed traffic, human drivers must also interpret and respond to autonomous vehicles (AVs), whose behavior, intent, recognizability, and social role may differ from those of human-driven vehicles. This paper presents a case-level evidence synthesis and develops a taxonomy of human driver errors and unsafe adaptations in mixed traffic, focusing on drivers of manually driven vehicles who interact with AVs as surrounding road users. The proposed taxonomy links AV-specific interaction factors, including recognizability, driving style, predictability, trust, social evaluation, infrastructure, penetration rate, and repeated exposure, to observable unsafe behavior and to the information-processing stages at which driver errors are inferred to occur. The synthesis suggests that many mixed-traffic errors are concentrated in the projection and decision stages, with two recurring pathways: safety-margin reduction caused by poor anticipation of AV behavior or excessive reliance on AV capability, and exploitation caused by intentional priority-taking or social devaluation of AVs. The taxonomy provides a structured basis for mixed-traffic safety analysis and assessment.

Keywords Driver error taxonomy, Behavioral adaptation, Autonomous vehicles, Human–AV interaction, Mixed traffic, Traffic safety

1 Introduction

Existing taxonomies of driver errors and violations were developed primarily for conventional traffic, where unsafe behavior is interpreted within a human–human interaction system involving human drivers, pedestrians, and cyclists [1, 2]. In mixed traffic, however, human drivers must interpret and respond not only to other human road users but also to AVs. This changes the interaction context because driving is not only a vehicle-control task but also a social interaction process involving communication, expectation, negotiation, and adaptation [3, 4]. Mixed traffic, therefore, does not simply reproduce familiar unsafe human behavior in a new environment. The presence of AVs may change how existing driver errors and violations occur by influencing how human drivers perceive AVs, anticipate their behavior, and decide whether to cooperate, yield, exploit, or maintain safety margins. The safe integration of AVs, thus, depends not only on autonomous vehicle-control performance but also on the behavioral responses of surrounding human road users.

Research on human factors in mixed traffic has expanded substantially, but it remains fragmented. One major research direction examines social interactions and the behavioral adaptation of human drivers around AVs in situations such as car-following, merging, overtaking, platooning, and intersection negotiation [5–7]. This literature indicates that human–AV interaction is rarely behaviorally neutral: human drivers may behave

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more cautiously in some contexts, but they may also accept smaller gaps, follow more closely, exploit perceived AV conservatism, or become uncertain about AV intent. A second strand of research focuses on automated driving system failures and broader mixed-traffic operation, including machine-side failures, interaction conditions between AVs and surrounding road users, driving characteristics in mixed traffic, operational impacts, and deployment strategies [8–10]. Existing work largely covers interaction effects, operational conditions, or AV-system failures rather than linking AV-specific conditions to human-driver errors, violations, and unsafe behavior that emerge in mixed traffic. As a result, the literature lacks a framework for systematically examining how AV-related behavioral adaptation emerges and how it influences mixed-traffic safety.

Accordingly, the objective of this study is to determine how AV-specific interaction factors reshape human-driver errors and violations and to organize these adaptations in a taxonomy. This paper addresses that gap by taking conventional driver error and violation taxonomies as the baseline and examining how they need to be interpreted or extended for mixed traffic. The central assumption is that mixed traffic may not create wholly new human driver error classes. Instead, it may reshape the conditions under which existing errors emerge by introducing *AV-specific interaction factors*, such as AV recognizability, AV driving style and rule-compliance, AV behavior predictability and intent ambiguity, trust in AV safety and automation capability, AV social status and moral accountability, AV-dedicated infrastructure, AV platoon exposure, AV penetration rate and traffic composition, and repeated AV exposure and learning.

Conventional taxonomies usually classify driver errors and violations by the driver process involved or by the observable failure. In mixed traffic, however, the same observable behavior may have different mechanisms depending on whether the interaction partner is a human-driven vehicle or an AV. For example, a short headway, small accepted gap, or delayed yielding response may reflect not only a conventional misjudgment, but also adaptation to perceived AV predictability, conservatism, recognizability, or trustworthiness.

Thus, the key question is not only whether human drivers adapt to AVs, but whether these adaptations alter the safety implications of human errors and violations relative to conventional traffic. A corresponding taxonomy enables comparison between conventional and mixed-traffic error patterns. It provides a structured basis for examining how behavioral adaptation to AVs may influence the occurrence and safety implications of human errors and violations, and helps identify the AV-specific interaction factors that contribute to unsafe behavior. The novelty of the study lies in the three-layer linkage between AV-specific interaction factors, inferred information-processing stage at which an error occurs, and observable behavior.

This paper makes two contributions. The first is a *taxonomy* that builds on established situation-awareness and error/violation frameworks and extends them to mixed traffic by linking AV-specific interaction factors to the information-processing stages at which driver errors occur and to the resulting observable behavior. The second is a case-level *evidence synthesis* of the current literature from which this taxonomy is derived, identifying the mixed-traffic safety problem mainly in the projection and decision stages. The case-level synthesis identifies two recurring pathways: projection-stage safety-margin reduction and decision-stage exploitation or social devaluation of AVs. Together, these contributions provide a structured basis for behavioral analysis and for the development of safety performance indicators in mixed traffic, supporting evidence-based approaches.

The paper proceeds as follows. Section 2 presents the methodology, with a component-flow overview of how Sects. 3 and 4 provide the input streams from which Sect. 5 derives the taxonomy. The paper concludes in Sect. 6 with implications, limitations, and directions for future work.

2 Methodology

This paper uses a narrative evidence synthesis to develop a taxonomy of human driver errors and unsafe adaptations in mixed traffic. In this paper, *unsafe adaptations* are defined as behavioral adjustments by human drivers in response to AV presence, behavior, or perceived capability that increase conflict risk. The scope of this synthesis is limited to human drivers of manually driven vehicles (MVs) interacting with AVs as surrounding road users. The focus is on interaction outside the AV control loop, where the human driver neither supervises nor shares control with the AV, but instead perceives, anticipates, and responds to the AV as another traffic participant. Across the literature synthesized in this paper, terminology for vehicle categories is not fully consistent: MVs, as human-driven vehicles, are alternatively referred to using abbreviations HVs or HDVs; AV is used for automated and autonomous vehicles; and CAV refers to connected and automated vehicles. Other road user groups, such as pedestrians, cyclists, and in-vehicle users of AVs, are outside the scope of the paper.

The synthesis follows a theory-informed, evidence-led approach. The information-processing stages were specified from established driver-error and situation-awareness frameworks. The AV-specific interaction factors build on constructs already established in the automation-trust and human-AV interaction literature and were refined against the reviewed evidence.

The observable-behavior groups were developed inductively from recurring patterns identified in the reviewed evidence.

Operationally, the synthesis comprised four steps: literature identification and eligibility screening, extraction of case-level evidence, coding of stage–factor–behavior attributes, and descriptive synthesis into the taxonomy. The overall procedure is summarized in Fig. 1, which also serves as a roadmap for the remainder of the paper. The flow representation borrows the identification, screening, eligibility, and inclusion sequence used in PRISMA reporting [11], but remains a roadmap and does not imply that this narrative evidence synthesis was conducted or reported as a PRISMA-compliant systematic review. The taxonomy is developed from two input streams. The first stream, reviewed in Sect. 3, is the conventional driver error and violation taxonomies. This stream plays a dual role: it identifies the gaps that motivate a mixed-traffic taxonomy, and it supplies the conceptual ingredients from which the taxonomy is built, namely the situation-awareness stages (perception, comprehension, projection) and the subsequent decision stage, together with the distinction between errors and violations. The second stream, reviewed in Sect. 4, is the empirical evidence on driver behavioral adaptation in mixed traffic, which provides the source material from which safety-relevant behaviors, inferred error mechanisms, and AV-specific interaction factors are extracted.

For the first input stream, established driver-error, error/violation, and situation-awareness frameworks were identified through targeted literature searching and backward and forward citation tracing from foundational and widely cited studies. Sources were selected based on their relevance to established classifications of driver errors and violations.

For the second input stream, a structured narrative literature search was conducted in Google Scholar, Transport Research International Documentation (TRID), Scopus, and Web of Science Core Collection, covering peer-reviewed journal and conference papers, published with no lower publication date limit through 17 April 2026, as well as databases of mixed traffic crashes. The search was supplemented by backward and forward citation searching from relevant reviews and primary publications. Eligible primary evidence included driving-simulator and field studies, naturalistic dataset and crash-report analyses, observational studies, driver self-reports, survey experiments and human-calibrated microsimulation. The search combined terms relating to AVs ("automated vehicle", "autonomous vehicle", "connected and automated vehicle") and MVs ("human driver", "manual vehicle", "manually driven vehicle", "manually controlled vehicle") and their "interaction" in "mixed traffic", in combination with terms relating to behavioral outcomes ("behavior" and "behavioral adaptation"). The core Boolean combination was (AV terms) AND (MV terms) AND (mixed-traffic/interaction terms) AND (behavior/behavioral-adaptation terms), with syntax adapted to each database. Retrieved publications were screened for relevance to AV–MV interaction and then assessed against the case-level eligibility criteria. The first-pass relevance screen was a subjective judgment by the authors, based on whether a publication addressed behavioral interaction between MVs and AVs in mixed traffic. Studies were eligible for case coding when they reported a safety-relevant behavioral outcome together with information about the AV-related condition that may have triggered or modulated it, such as AV's behavior, appearance, or interaction context.

Reviews or conceptual analyses, and aggregate crash or simulation results from which the human behavior in relation to interaction with an AV could not be recovered were excluded from case coding. Real-world evidence from the Waymo Open Dataset [12] and California DMV collision reports [13, 14] was also examined and case-coded only when it satisfied the same behavioral-linkage criteria. More than 60 publications and data sources were assessed. 37 of them met the eligibility criteria, yielding

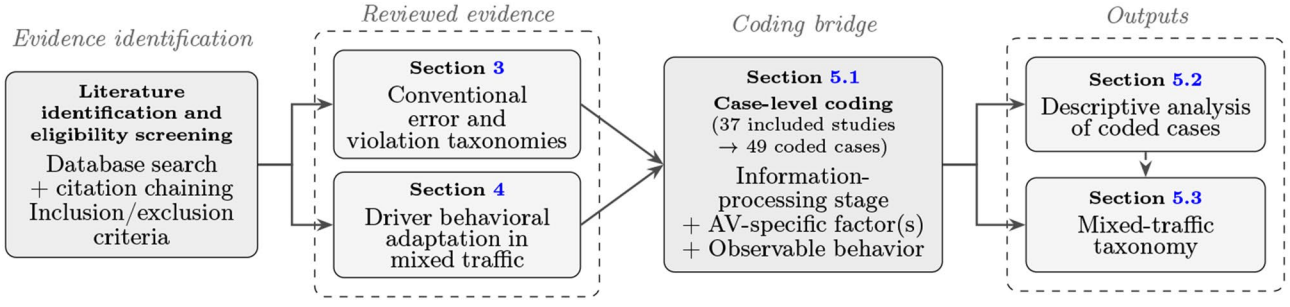


Fig. 1 Roadmap and component-flow overview of the synthesis. Literature identification and eligibility screening provide the literature foundation for the two reviewed input streams: targeted searching and citation tracing for the conventional driver error and violation taxonomies in Sect. 3, and structured database search with explicit eligibility criteria for the behavioral-adaptation evidence in Sect. 4. These streams are linked in Sect. 5.1 through eligible studies case-level coding, which records the information-processing stage, AV-specific interaction factors, and observable behavior for each case. The coded cases are then summarized descriptively in Sect. 5.2 and synthesized into the proposed mixed-traffic taxonomy in Sect. 5.3

49 evidence cases. A single study could contribute more than one case when it reported distinct interaction mechanisms or safety-relevant outcomes. The concentration of eligible studies in recent years – 29 out of the 37 studies were published from 2021 to 2026 – reflects the recent expansion of empirical mixed-traffic research and publicly available AV and mixed-traffic crash datasets. Not every included study is discussed individually in Sect. 4; the complete set of 37 studies and 49 coded cases is reported in Additional file 1, "Case-Level Synthesis of Human-Driver Errors and Unsafe Adaptations in Mixed Traffic".

Evidence from each study was coded using the three attributes shown in Fig. 1: the observable driver behavior that compromises traffic safety, the information-processing stage at which the error is inferred to occur, and the AV-specific interaction factors that appeared to trigger or modulate the behavior. These attributes, together with the detailed factor codes and observable-behavior groups, are defined in Sect. 5, where they are used as the building blocks of the taxonomy.

3 Driver error taxonomies in conventional traffic

This section reviews driver error and violation taxonomies developed for conventional traffic. It serves two purposes in the synthesis. First, it identifies the gaps that motivate a dedicated mixed-traffic taxonomy. Second, it supplies the conceptual ingredients from which that taxonomy is built: the situation-awareness stages and the distinction between errors and violations that later structure the coding of mixed-traffic cases.

Human error remains a major issue in road safety, but human driver errors are less systematically classified than errors in some other safety-critical domains. Driver error refers to failures in the driver's cognitive processing, situation assessment, decision-making, or action execution [1, 15]. It describes *why* the driver failed to react appropriately in a traffic situation. Violations are conceptually distinct from unintentional errors. They involve deviations from accepted rules or legal requirements, although they may be deliberate or unintentional [15, 16]. This distinction matters because the visible outcome may be similar even when the underlying mechanism differs. A vehicle may travel too fast because the driver intentionally violates the speed limit, fails to notice a change in the speed limit, misjudges the safe speed for the current road conditions, or executes the intended control action poorly. From a legal perspective, this distinction is also not equivalent to safety status. A behavior may be safe and legal, unsafe but legal, safe but illegal, or unsafe and illegal. For example, driving at the posted speed limit on a snow-covered road may be legal but unsafe, whereas an illegal maneuver on an empty road may be unlawful but not immediately safety-critical [17]. A useful driver error taxonomy should therefore distinguish the cognitive or behavioral mechanism from the legal status of the action and indicate the information-processing stage and psychological mechanism from which the observed action may arise.

A widely used distinction in driving research derives from the broader human-error literature, including slips, lapses, mistakes, and violations framework proposed by Reason et al. [15]. Slips are usually associated with attentional or execution failures, lapses with memory failures, and mistakes with poor planning, judgment, or decision-making.

Stanton and Salmon [1] provide one of the most comprehensive attempts to apply human error taxonomies to driving and propose a generic driver error taxonomy based on the dominant psychological mechanisms involved. These mechanisms include perception, attention, situation assessment, planning and intention, memory and recall, and action execution. The taxonomy is accompanied by a causal-factor taxonomy covering road infrastructure, vehicles, other road users, and environmental conditions. This is consistent with a systems view of road safety, in which error is shaped by the driver, vehicle, road environment, traffic rules, other road users, and technology design. At the same time, this taxonomy remains a generic synthesis rather than a fully operational taxonomy. The authors note that the proposed taxonomies should inform data collection and require validation and refinement through further study. Although the taxonomy is organized around psychological mechanisms, it does not explicitly follow a sequential model of human information processing in which situation awareness precedes decision-making and action. The psychological mechanisms are therefore not arranged as a staged model that separates perception of traffic elements, comprehension of their meaning, projection of their near-future status, decision selection, and action execution.

Situation awareness offers a useful structure for distinguishing visible driver behavior from the underlying information-processing error. Following Endsley's three-level model, situation-awareness failures can be understood as failures to perceive relevant information, comprehend the situation, or project how the situation will develop [18, 19]. Although rooted in aviation research, these levels map well onto driving [1, 20]. Level 1 concerns the perception of relevant traffic elements, such as vehicles, pedestrians, traffic signals, lane markings, and speed changes. Level 2 concerns the comprehension of what these elements mean for the current traffic situation, such as right-of-way, conflict risk, or another road user's intention. Level 3 concerns the projection of how the situation will develop, such as whether a gap will remain available, whether a lead vehicle will brake, or whether another road user will enter the conflict

area. Situation awareness forms the basis for decision-making, which refers to the selection of an action strategy, including yielding, merging, overtaking, tailgating, or deliberately taking priority. A driver may make a correct decision for the situation as understood, while the underlying situation model is wrong. For driving, the implication is that taxonomy development should not stop at visible behavior. It should also attempt to identify whether the observable behavior was caused by failure to perceive, understand, predict, decide, or execute.

Naturalistic driving studies provide an empirical basis for classifying errors in real traffic, but they also show the limits of available data. Khattak et al. [2] developed a taxonomy of driving errors and violations using SHRP2 Naturalistic Driving Study (NDS) data from Transportation Research Board of the National Academies of Science [17]. Their taxonomy uses the evolution of driving errors and violations framework, with perception–recognition–decision–reaction phases, while treating physical-condition errors and experience or exposure errors separately. A major contribution of this work is its connection to real-world pre-crash data, including crashes and near-crashes. Their analysis reports that human errors and violations were identified in most crashes, with recognition and decision errors being the most frequent categories, providing an important empirical baseline for conventional traffic.

However, the use of NDS data also illustrates a central methodological problem for driver error taxonomies. NDS data are based on instrumented vehicles, sensor measurements, video, and data-reduction protocols. This allows observation of speed, acceleration, braking, lane position, eye glance, secondary tasks, and the visible sequence of safety-critical events. Yet the underlying cognitive process is not directly observable. This limitation also arises in conventional crash databases, where human-related and aberrant-behavior variables can be difficult to reconstruct reliably from accident records [21]. A data reductionist may observe that a driver did not react, ran a red light, drove slowly, exceeded a safe speed, misjudged a gap, or failed to signal correctly, but the internal reason may remain unclear. Some NDS examples suggest relatively clear mechanisms, such as a driver who did not notice a traffic sign, saw a traffic signal but purposefully disregarded it, misunderstood right-of-way, or drove in an intentionally aggressive manner. Other examples are more ambiguous, such as proceeding through an uncontrolled intersection in an unsafe manner, driving below the speed limit when higher speeds were appropriate, or exceeding the posted speed limit in a work zone. Such events can be coded as errors or violations, but the available data may not fully distinguish perception failure, comprehension failure, wrong projection, poor decision, intentional violation, or contextual uncertainty.

This limitation is a consequence of how safety-compromising events are observed and coded. Observable behavior is the visible endpoint of a longer cognitive process. Internal processes such as perception, recognition, and decision-making are only partly recoverable from event video and vehicle kinematics. Khattak et al. [2] partially address this by using narratives and secondary-task variables when driver behavior variables are missing, but some events remain difficult to classify because video views, lighting, visual obstructions, or limited perspective restrict what can be known. Therefore, naturalistic data are valuable for identifying what happened, but they are less reliable for determining why it happened unless combined with richer contextual, behavioral, and subjective information.

Existing driver error taxonomies provide useful foundations for classifying human driver errors in conventional traffic. For the present purpose, however, they need to be extended with an interaction layer that captures how AV-specific conditions may shape the perception, interpretation, anticipation, and decisions of surrounding human drivers. Stanton and Salmon [1] provide a strong human-factors structure and a broad systems perspective, but their taxonomy is not organized as a staged situation-awareness–decision–action framework and was not developed for mixed traffic. Khattak et al. [2] provide a data-linked taxonomy based on NDS crashes and near-crashes, but the mapping from observed driving behavior to underlying error mechanisms can be uncertain, and the taxonomy was not developed to capture mixed-traffic interactions involving AVs. Traffic-situation taxonomies for AV-human road user interaction also exist [8], but these describe interaction situations rather than classifying human driver errors in mixed traffic. Similarly, recent mixed-traffic studies and reviews examine interaction, trust, behavioral adaptation, and safety-relevant behavior, but they do not provide a taxonomy of human driver errors caused, shaped, or modulated by interaction with AVs.

The research gap, therefore, leaves two related issues unresolved. First, there is still no generally accepted framework that systematically links observed behavior to failures occurring at earlier information-processing stages, such as perception, comprehension, projection, or decision-making. Second, to the best of our knowledge, there is no established human driver error taxonomy for mixed traffic that accounts for AV-specific interaction factors such as AV recognizability, driving style, predictability and intent ambiguity, trust, social and moral evaluation, dedicated infrastructure, penetration rate, and repeated exposure. This section thus contributes both halves of its dual role: the two gaps above, which motivate

the taxonomy, and the situation-awareness–decision scaffold and error/violation distinction, which become the ingredients used to structure it. The remaining ingredient, the AV-specific interaction context in which these failures occur, is supplied by the behavioral-adaptation evidence reviewed next in Sect. 4.

4 Driver behavioral adaptation in mixed traffic

This section reviews the empirical evidence on how human drivers adapt their behavior in mixed traffic. It is the second input stream in Sect. 2 (Fig. 1): the source material from which the safety-relevant behaviors, inferred error mechanisms, and AV-specific interaction factors used in the taxonomy are extracted.

Predicting how a human road user will react in a particular traffic situation remains a significant challenge. Human drivers may exhibit rule noncompliance, delayed responses, signal misperception, communication failures, distraction, and attentional failures. They may stop at an unintended location, fail to stop, accelerate unexpectedly, brake abruptly, ignore speed limits, or overlook messages about restrictions and instructions. In conventional traffic, such manifestations are usually interpreted within a human–human norm system and are commonly linked to distraction, fatigue, misjudgment, aggression, or violations. In mixed traffic, many of the same observable behaviors can occur, but their triggers and interpretation may differ because human drivers interact with AVs whose behavior, intent, capability, and social role may be perceived differently from those of MVs. Consequently, human road users adapt their behavior to the presence of AVs in traffic.

Behavioral adaptation is treated as the process through which drivers adjust their actions, expectations, and interaction strategies in response to changes in the road traffic system. The OECD definition describes behavioral adaptation as unintended behavioral change following the introduction of a new system [22], while later work broadened the concept to include any driver response to changes in the road traffic system [23].

Classical road-safety research provides a broader basis for interpreting such adaptation. Wilde [24]'s risk-homeostasis framework and Elvik [25]'s work on behavioral adaptation show that perceived changes in safety can induce compensating changes in driver behavior, while Kononov et al. [26] demonstrate that safety outcomes depend on traffic-flow conditions such as speed, density, and exposure.

The concept of behavioral adaptation in the context of AVs has gained attention in human factors studies, particularly concerning the behavioral changes of drivers in MVs interacting with AVs. This perspective is relevant, as mixed traffic is not only a technical coexistence problem but also a social-interaction problem. Roads are social spaces in which drivers coordinate through formal rules, informal norms, motion cues, signals, and implicit expectations [6, 7, 27]. AVs may disrupt this interaction system because they are often expected to be conservative, rule-compliant, and yielding, while human drivers may be uncertain about their intent, limitations, and likely responses. Existing empirical research suggests that human drivers behave differently around AVs than around MVs. Findings vary across scenarios and methods, but recent reviews indicate mixed evidence with an overall negative trend that humans may exploit AVs or interact with them less cautiously, while AV behavior may also induce behavioral changes that create new challenges for human drivers [7]. At the same time, mixed-traffic outcomes are not uniformly negative; some studies report smoother car-following, lower volatility, or no significant difference [28, 29], indicating that behavioral adaptation depends on interaction context, AV driving style, recognizability, traffic composition, and prior experience.

To make this link useful for traffic safety, behavioral adaptation must be connected to measurable indicators. The outcomes reviewed here are grouped into observable behavioral indicators and subjective indicators. Behavioral indicators are direct manifestations of adaptation in traffic interaction. These include time headway (THW), distance headway, critical gap and gap acceptance, speed choice, acceleration and braking patterns, lane-change timing, merging behavior, yielding, and overtaking distance.

Subjective indicators capture how road users evaluate interaction with AVs. These include trust, comfort, perceived safety, acceptance, workload, anger, and willingness to share the road. Although subjective measures do not directly indicate safety, they shape behavioral choices and help explain why similar traffic situations produce different observable responses. Trust is particularly central: higher trust may increase willingness to follow or merge in front of AVs, but it may also reduce following distance or encourage overreliance if drivers overestimate AV capabilities [30–32]. Perceived safety and comfort are similarly important in scenarios involving AV recognizability, external human–machine interfaces, and AV driving style. Defensive AV behavior may be perceived as more trustworthy and predictable [33], but overly cautious or slow AV behavior may also increase irritation or discomfort and provoke more assertive or aggressive interaction [34–36].

The reviewed evidence can be interpreted through recurring interaction contexts. In car-following, mixed traffic primarily affects recognition of AV behavior and proximity to AVs. Field and naturalistic data studies, including analyses based on real-world datasets such as the Waymo Open Dataset, suggest that MV drivers may

behave differently when following AVs than when following other MVs [12, 37]. Some studies report lower driving volatility, smoother speed adjustments, and improved safety indicators when MVs follow AVs [29, 37, 38]. Other work shows that the effect depends on drivers' subjective trust and whether the AV is recognizable. When AVs are distinguishable, drivers who believe in AV technology may follow more closely, whereas AV-skeptical drivers may maintain larger gaps [28]. Crash-report studies, including analyses of California DMV AV collision reports, provide a safety-relevant counterpart to these behavioral findings: in AV crash reports, rear-end collisions were frequent, most rear-end cases involved the AV being struck by another vehicle [13, 14, 39, 40], and rear-end collisions were positively associated with AVs being stopped, slowing, yielding, or waiting [40]. Similarly, conventional-vehicle drivers involved in AV crashes were more often coded with unsafe speed and following too closely than drivers in comparable conventional-vehicle crashes, and rear-end crashes were more frequent in AV crashes than in conventional-vehicle crashes at the same locations [39]. These findings indicate that following behavior is not determined by AV presence alone, but by the interaction between AV motion, driver expectations, trust, and the human driver's ability to anticipate AV deceleration, stopping, or yielding. Complementary real-world evidence comes from analyses of the Waymo Open Dataset and California DMV AV collision reports, which provide observed AV–MV trajectories and pre-crash patterns outside controlled experiments [12–14].

Merging and gap acceptance show a more direct link between AV-specific expectations and potentially risky adaptation. Several studies suggest that drivers may accept smaller gaps or merge more assertively in front of AVs, especially when they believe AVs are programmed defensively [5, 41]. This should not necessarily be interpreted as a conventional merging error. In mixed traffic, the behavior may arise from an expectation that the AV will yield or brake reliably, due to an overestimation of AV cautious driving behavior. This phenomenon, known as automation complacency [30, 41], raises concerns about potential traffic hazards if drivers overestimate AV capabilities and fail to account for real-world limitations.

Reddy et al. [42] further showed that drivers adjust gap acceptance according to both AV recognizability and driving style: recognizable aggressive AVs led to larger critical gaps, whereas non-recognizable aggressive AVs led to smaller critical gaps. These findings indicate that recognizability and driving style are important contextual factors in a mixed-traffic taxonomy. A short critical gap may represent misjudgment, but it may also represent strategic exploitation of perceived AV conservatism.

Lane changing and overtaking near AVs or AV platoons provide another example of context-dependent adaptation. AV platoons and dedicated AV lanes can alter local traffic norms because they introduce vehicles with shorter headways, stricter speed control, or more uniform movement. Gouy et al. [43] found that human drivers reduced their own headways when driving beside AV platoons with short headways. Similar effects were reported in studies of dedicated lanes, where MV drivers near AV platoons followed more closely or accepted shorter merging gaps [44, 45]. Crash-report evidence also suggests that lane-changing and overtaking interactions can be safety-critical: in crash reports, passing or overtaking of a CAV by a following conventional vehicle was identified as a movement combination that may lead to crash occurrence, and sideswipe crashes accounted for a substantial share of CAV-involved crashes [46]. Such AV movement patterns may generate a contagion effect among human drivers. Features intended to improve traffic flow, such as stable close-headway platooning, may therefore create unsafe adaptation if human drivers imitate these patterns without the same automation capabilities [43–45, 47].

The penetration rate of AVs changes the interaction context from a single encounter to a traffic-composition effect. As AV penetration increases, human drivers may encounter AV behavior more frequently and may adapt to the emerging local norm. Simulator studies suggest that increasing AV penetration can lead drivers to adopt behaviors closer to those of AVs, including shorter headways, smaller following distances, and less variable speed [47]. Stange et al. [48] also observed that higher CAV presence may reduce manual drivers' average speed while increasing safety-critical interactions with lead vehicles. The mixed-traffic literature similarly identifies penetration rate and safe mixed-traffic operation as central research topics [9]. At higher AV penetration rates, drivers may adapt to the traffic stream itself, and errors may emerge from altered expectations about normal speed, headway, yielding, or lane discipline. Microsimulation provides a complementary evidence source for mixed-traffic conditions by allowing changes in car-following, lane-changing, traffic conflicts, and other safety-relevant interaction measures to be examined under different AV assumptions and penetration levels [47, 49, 50].

Repeated exposure and learning further complicate the effect of AV–MV interactions. Initial encounters of MVs with AVs may be dominated by uncertainty, novelty, curiosity, or mistrust, whereas repeated encounters may produce overconfidence or strategic exploitation. Stange et al. [48, 51] found that human drivers could distinguish CAVs from MVs using behavioral cues, and that external Human-Machine Interfaces (eHMIs) improved recognition but did not necessarily change subjective ratings or driving behavior. Other studies show that behavior may change across repeated interactions as drivers learn the

properties of AVs and adjust their speed, headway, or reaction time [51]. This suggests that mixed-traffic error patterns may evolve over time.

For the development of a driver error taxonomy in mixed traffic, the evidence indicates that behavioral adaptation is shaped by both observable interaction conditions and subjective evaluations. Behavioral indicators such as headway, speed, acceleration, braking, gap acceptance, and lane-change timing provide evidence of adaptation, whereas subjective indicators such as trust, comfort, perceived safety, anger, and acceptance help explain why adaptation occurs. Their interpretation depends on the interaction context. A short headway behind an MV may be classified within conventional car-following errors, whereas a short headway behind a recognizable AV may reflect trust, exploitation, imitation, or perceived predictability. Similarly, a risky merge in front of an AV may reflect a conventional gap-acceptance error, but it may also be an AV-specific behavioral adaptation based on the expectation that the AV will yield.

Therefore, for the purpose of classifying unsafe adaptations in mixed traffic, it is useful to record not only the observed behavior but also the AV-related interaction context that appears to have triggered or modulated it. Mixed traffic introduces new sources of uncertainty, including AV intent ambiguity, differences between MV and AV driving styles, inconsistent AV behavior across manufacturers, recognizability, external communication, penetration rate, and learning over repeated exposure [6, 7, 52]. These factors do not replace conventional driver error mechanisms; rather, they reshape the conditions under which situation awareness, decision-making, and action-execution failures occur. Behavioral adaptation, therefore, provides the link between conventional driver-error taxonomies and a mixed-traffic taxonomy.

The evidence reviewed in this section further suggests that subjective and contextual indicators may influence the internal information-processing stages that precede observable behavior. In particular, they may affect situation awareness at the levels of comprehension and projection, as well as subsequent decision-making. When these internal processes are inaccurate, they may emerge as observable unsafe behavior, such as short headways, unsafe gap acceptance, late braking, abrupt lane changes, or non-yielding. Such behaviors are therefore best interpreted as visible outcomes of earlier cognitive processes rather than as self-explanatory error categories. Depending on intention and context, the same observable action may result from a performance error, a decision error, or a violation.

5 Proposed taxonomy for mixed traffic

This section presents the output of the synthesis, combining the two input streams, presented in Sect. 2 (Fig. 1). It draws the situation-awareness–decision scaffold and the error/violation distinction from literature in Sect. 3, and the AV-specific interaction factors and safety-relevant behaviors from the evidence reviewed in Sect. 4. The proposed taxonomy links AV-specific interaction factors that trigger or modulate human driver behavior to the resulting observable behavior and to the information-processing stage at which the driver error is inferred to occur. The subsections follow the coding step and its two outputs, shown in Fig. 1: Sect. 5.1 defines the case-level coding scheme, Sect. 5.2 reports the descriptive analysis of the coded cases, and Sect. 5.3 presents the resulting taxonomy structure.

The taxonomy was developed from the available evidence reported in each study, including observable behavior, experimental conditions, traffic context, and the explanation of why the behavior occurred. The resulting set of coded cases was then summarized descriptively to assess how cases were distributed across information-processing stages, observable behavior groups, and AV-specific interaction factors. The final taxonomy is therefore based on conceptual links between the attributes, while the descriptive analysis is used to identify the dominant patterns represented in the reviewed evidence.

5.1 Case-level coding and taxonomy attributes

The taxonomy is organized around case-level evidence units. In this paper, a *case* refers to a coded observation extracted from the reviewed literature in which a human driver's unsafe behavior, inferred error mechanism, and AV-related interaction context can be linked. A case, therefore, connects what the driver did, why the behavior is interpreted as unsafe, and which AV-specific condition appeared to shape it. The inferred driver-error mechanism ('why the driver failed') was derived from the explanation supported by each study, using the reported experimental condition, behavioral outcome, traffic context, driver self-report where available, and the authors' interpretation of the observed effect.

Each case was coded using three taxonomy attributes: the information-processing stage at which the driver error is interpreted to occur, the AV-specific interaction factors that trigger or modulate the behavior, and the resulting observable behavior. This three-part structure was used because observable behavior alone is not self-explanatory in mixed traffic. The same visible action, such as accepting a small gap, may reflect a generic gap-judgment error in one context, but may reflect perceived AV predictability, trust in AV braking, or exploitation of expected AV conservatism in another. Therefore, each attribute captures a different level of explanation.

Observable behavior records what was visible in traffic interaction; the information-processing stage records the inferred cognitive or decision point at which the error occurred; and the AV-specific interaction factor records the mixed-traffic condition that may have shaped the behavior. The coding structure, therefore, links context, inferred mechanism, and observable safety-relevant consequence in a single case-level unit.

Information-processing stage at which error occurs: Each case is assigned to the information-processing stage at which the driver error is interpreted to occur: perception, comprehension, projection, or decision. *Perception* was coded when the primary inferred mechanism concerned detecting or noticing relevant traffic information, or perceptual contagion and calibration to the observed traffic patterns, before interpretation of their situational meaning. *Comprehension* was coded when the information was perceived, but its meaning for the current traffic situation was misinterpreted. *Projection* was coded when the primary inferred failure concerned anticipating what the AV would do next or how the situation would develop. *Decision* was coded when the situation was sufficiently understood, but the driver selected an unsafe action, including an assertive or strategic response. Where more than one stage was plausible, the case was assigned to the stage most directly supported by the reported mechanism, using the experimental condition, behavioral outcome, traffic context, driver self-report where available, and the study authors' interpretation.

AV-specific interaction factors: The contextual and subjective indicators assigned to each case capture AV-specific conditions that may shape human-driver behavior. These factors build on constructs already established in the automation-trust and human–AV interaction literature (e.g., recognizability, trust, driving style) and were refined against the reviewed evidence: they were retained when they recurred across the mixed-traffic literature, described an AV-specific interaction condition, and could be linked to an observable safety-relevant outcome in the coded cases. During synthesis, initially identified factors and behaviors were iteratively consolidated where they represented the same underlying interaction condition. The resulting factors were then applied consistently across the complete case set for the descriptive analysis. They are therefore not intended to exhaust all possible mixed-traffic variables, but to capture the recurring AV-specific triggers and modulators needed to explain the unsafe adaptations identified in the evidence base. After systematizing the available evidence, the following factors were identified and used across the cases:

ID-CUE – AV recognizability and identity cues
STYLE – AV driving style and rule compliance
PRED – AV behavior predictability and intent ambiguity
TRUST – Trust in AV safety and automation capability
SOC-MOR – AV social status and moral accountability
INFR – AV-dedicated infrastructure
PLT – AV platoon exposure
PEN – AV penetration rate and traffic composition
EXP – Repeated AV exposure and learning

The factors were coded as multi-label attributes rather than as mutually exclusive or collectively exhaustive categories. In many cases, unsafe behavior emerges from a combination of factors. For example, a driver may accept a smaller gap in front of an AV because the AV is recognizable, is expected to behave defensively, and is trusted to brake reliably. Similarly, aggressive behavior toward an AV may be shaped by AV driving style, perceived lack of social accountability, and the expectation that the AV will avoid collision even when the human driver behaves assertively.

Observable behavior: Observable behaviors are recorded at the specific level supported by the reviewed evidence, such as short time headway, short following distance, small accepted gap, delayed yielding, right-of-way violation, reckless cut-in, tailgating, or brake-checking. To make these case-level behaviors comparable across studies, each behavior is also assigned to one broader behavior group. These groups are deliberately behavioral rather than legal categories and describe what can be observed in traffic interaction. Four behavior groups are used in the synthesis and coded:

Reduced safety margins (SM) include behaviors that reduce longitudinal, lateral, temporal, or standstill buffers during car-following, stopping, overtaking, or passing.

Right-of-way and yielding behavior (ROW) includes unsafe or inefficient priority negotiation, such as right-of-way violations, refusal to yield, delayed yielding, late braking when yielding, or unnecessary waiting in response to uncertainty.

Gap and lane-change behavior (GLC) includes gap acceptance, merging, cut-in, lane-changing, overtaking, and returning to lane when these behaviors are shaped by AV interaction.

Aggression or exploitation (AGG) includes intentional or strategic behaviors directed at AVs, such as bullying, tailgating as pressure, brake-checking, racing, blocking, or exploiting a conservative AV response.

Table 1 summarizes, by observable behavior group, the 49 coded cases reported in full in Additional file 1, together with the dominant information-processing stage, the most frequent AV-specific interaction factors, and studies contributing to each group.

To illustrate the coding logic, one case involved a human driver accepting smaller critical gaps when merging in front of an AV than in front of an HDV (Table 2). The observable behavior was coded as *small accepted/critical gap in front of an AV*, which belongs to the *gap*

Table 1 Overview of coded cases by observable behavior group

Behavior group	n	Dominant stage	AV-specific factors	Contributing studies
SM (reduced safety margins)	21	Projection (16)	PRED, TRUST, STYLE	[5, 12–14, 28, 32, 36, 39, 43–45, 48, 51, 53–60]
GLC (gap / lane-change behavior)	12	Projection (8)	PRED, TRUST	[5, 41, 42, 44, 61–64]
ROW (right-of-way / yielding)	9	Decision (6)	PRED, STYLE, SOC-MOR	[56, 63, 65–70]
AGG (aggression / exploitation)	10	Decision (10)	SOC-MOR, STYLE, ID-CUE	[34–36, 63, 71, 72]

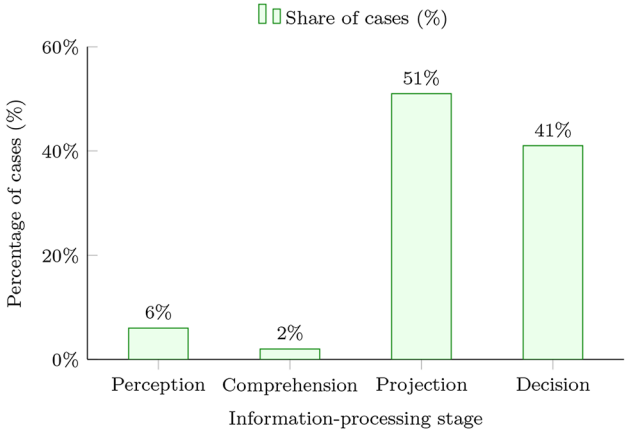


Fig. 2 Percentage of cases by information-processing stage

and lane-change behavior group. The inferred error was interpreted at the projection stage because the driver appeared to expect the AV to detect the merge and maintain safety from behind. The AV-specific interaction factors were coded as PRED and TRUST, reflecting the expected predictability of the AV response and trust in the AV's sensing and braking capabilities.

The full case-level synthesis is provided in Additional file 1. The table in the file presents one case per row, with the information-processing stage, AV-specific interaction factors, inferred driver error, observable behavior prefixed by its behavior group, and situation example reported in separate columns. In total, 49 cases were collected and synthesized. No separate cases were withheld for independent validation; therefore, the taxonomy should be regarded as an evidence-informed conceptual synthesis whose external validity requires testing against additional mixed-traffic evidence.

5.2 Descriptive analysis of coded cases

The coded cases were summarized descriptively to identify the main patterns in the evidence base and to support the development of the taxonomy. The analysis shows how the reviewed cases are distributed across information-processing stages, observable behavior outcomes, and AV-specific interaction factors. Because observable behaviors and AV-specific interaction factors were coded as multi-label variables, the stage distribution is reported as a percentage of cases, whereas the behavior and interaction-factor distributions are reported as percentages of coded occurrences. These patterns are then used to identify the most frequently observed mechanisms and recurring interaction pathways represented in the taxonomy. These distributions should be interpreted as descriptive, interpretive summaries of the reviewed cases. The assignment of information-processing stage, interaction factors, and behavior group involves interpretive judgment and was not subjected to a formal coding-reliability assessment. The counts indicate patterns represented within this synthesis rather than reliability-validated coding results.

The distribution of the 49 coded cases suggests that mixed-traffic human driver errors and unsafe adaptations are concentrated mainly in the projection and decision stages, as shown in Fig. 2. Within the current evidence base, this suggests that many mixed-traffic safety problems concern the driver's anticipation of what the AV will do and the driver's subsequent choice of action. Perception and comprehension cases are less frequent, but they remain important because AV recognizability, external communication, platooning, and dedicated

Table 2 Illustrative example of case-level coding

Information processing stage at which error occurs	AV-specific interaction factor(s) – trigger	Human driver error – why the driver failed	Observable behavior	Situation example
Projection	PRED; TRUST	Expects the AV to detect the merge and maintain safety from behind. Overtrusts the AV's sensing and braking, expecting it to compensate if the human driver merges into a smaller gap	*GLC*: Small accepted/critical gap in front of an AV	Driver **accepts smaller critical gaps when merging in front of an AV** than in front of an HDV. In the field test, the indicated critical gaps were significantly smaller in AV-interaction scenarios than in HDV-interaction scenarios, especially after positive information about AV behavior was provided. [5]

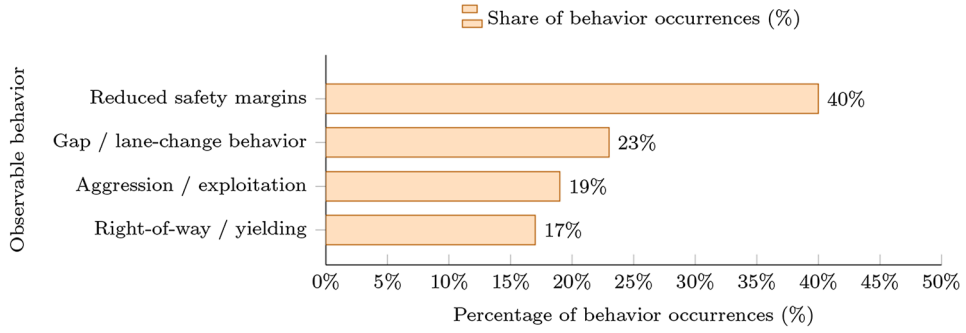


Fig. 3 Distribution of observable behavior groups. Behaviors are counted as multi-label outcomes, so one case may contribute to more than one behavior group. The 49 coded cases produced 52 behavior-group assignments. Percentages are calculated relative to the total number of coded behavior occurrences and represent each group's share of the 52 assignments. The displayed percentages sum to 99%, because values are rounded to whole percentages

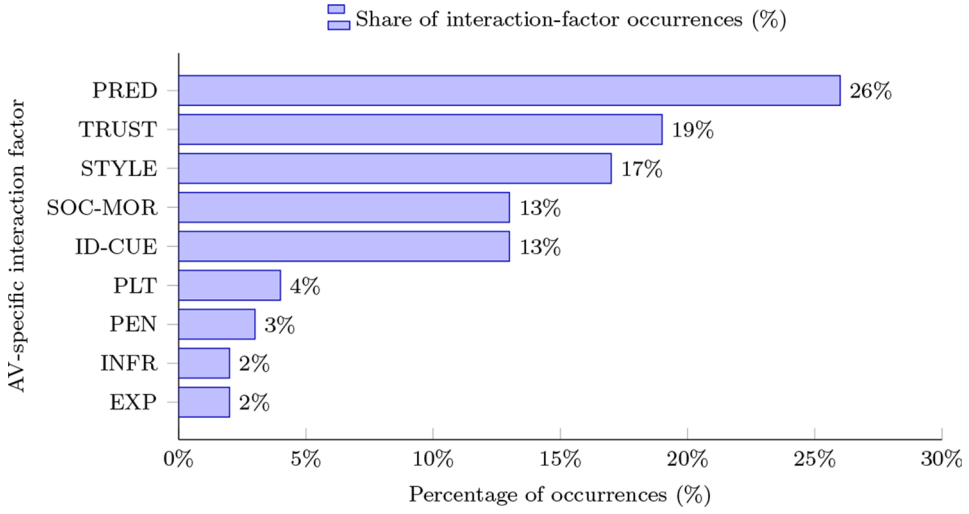
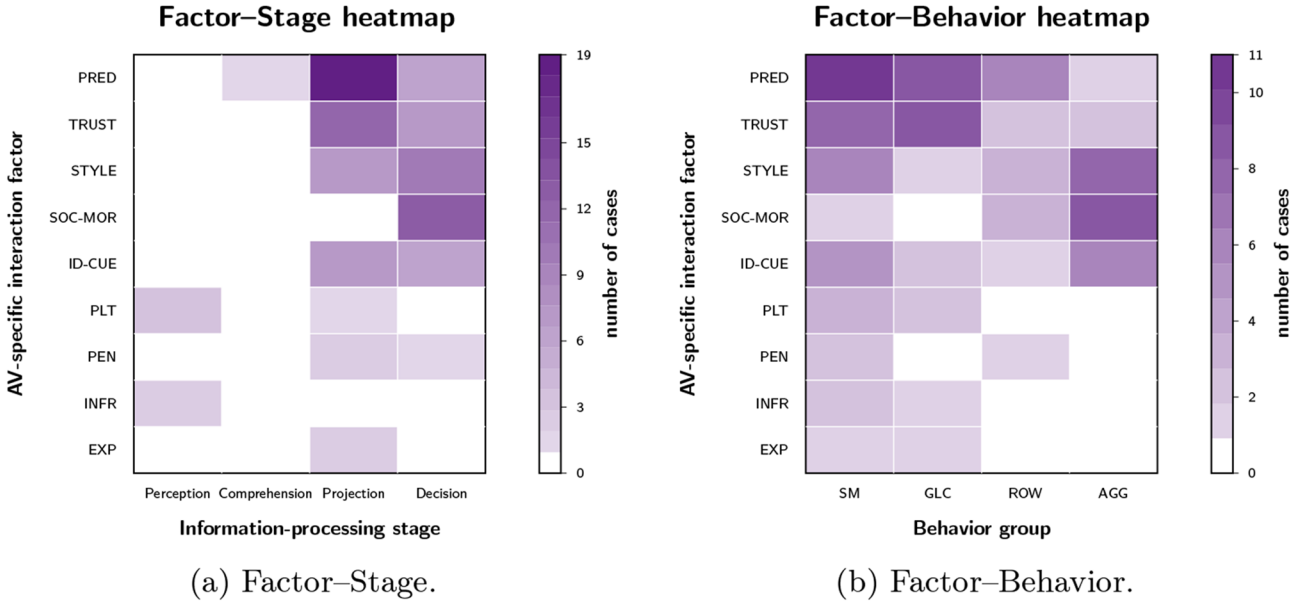


Fig. 4 Frequency of coded AV-specific interaction factors across identified cases. Interaction factors are counted as multi-label variables, so more than one factor may contribute to one case. The 49 coded cases produced 99 interaction-factor assignments. Percentages are calculated relative to the total number of coded interaction-factor occurrences and represent each factor's share of the 99 assignments. The displayed percentages sum to 99%, because values are rounded to whole percentages

infrastructure may alter how drivers detect and interpret the traffic situation.

The observable behavior groups in Fig. 3 show that the largest share of coded behavior occurrences concerns reduced safety margins. This group accounts for 40% of behavior occurrences and includes short THW, close following, smaller standstill gaps, reduced overtaking buffers, and related following-control risks. Gap and lane-change behavior accounts for 23% of behavior occurrences, followed by aggression or exploitation at 19%, and right-of-way or yielding behavior at 17%. This pattern is consistent with two broad interpretations. First, a major part of the mixed-traffic problem represented in the coded evidence is projection-stage safety-margin reduction: human drivers may overtrust AV capability, treat AV behavior as predictable, or expect the AV to compensate for unsafe human actions. Here, overtrust refers to reliance on AV capability that exceeds the system's actual limits, leading drivers to underestimate risk. This pathway may produce shorter headways, smaller gaps, closer following, and reduced overtaking margins. Second, another important part of the problem is decision-stage exploitation or social devaluation: drivers may treat AVs as lower-priority, less socially protected, or less risky targets and safer-to-bully interaction partners. This may produce aggressive decisions, reduced yielding, brake-checking, racing, close cut-ins, and intentional exploitation of conservative AV behavior.

The interaction-factor distributions further clarify the role of AV-specific conditions. As shown in Fig. 4, PRED is the most frequently represented interaction factor, accounting for 26% of coded interaction-factor occurrences. This is followed by TRUST at 19%, STYLE at 17%, SOC-MOR at 13%, and ID-CUE at 13%. The remaining factors are less frequent.



(a) Factor–Stage. (b) Factor–Behavior.

Fig. 5 Heatmaps of AV-specific interaction-factor identified across taxonomy dimensions: (**a**) by information-processing stage, (**b**) by observable behavior group. Counts reflect multi-label coding; a case may contribute to multiple factors. (**a**) Factor–stage. (**b**) Factor–behavior

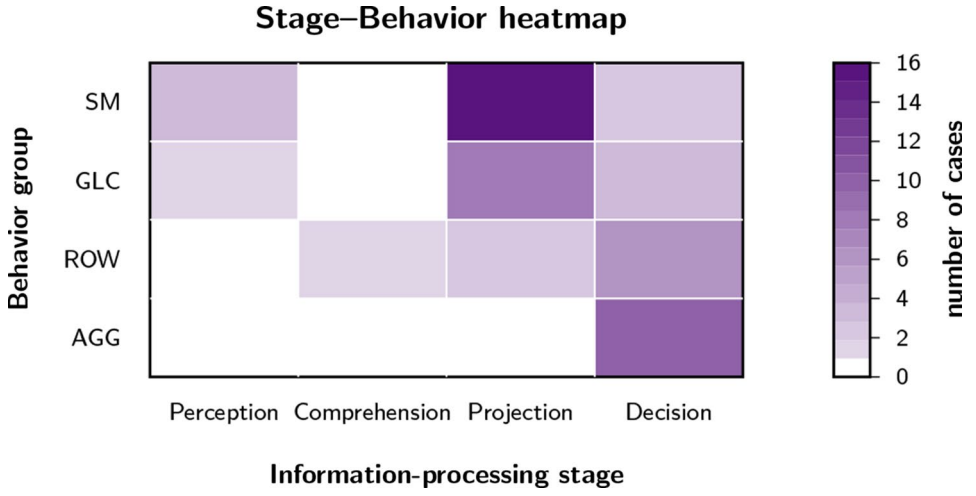


Fig. 6 Stage–behavior heatmap showing the distribution of coded observable-behavior occurrences within each information-processing stage. Behaviors are treated as multi-label outcomes, so a single case may contribute to more than one behavior group

Figure 5 shows how these factors are distributed across information-processing stages and behavior groups. In the Factor–Stage heatmap (Fig. 5a), PRED and TRUST are concentrated mainly in projection-stage cases, STYLE and ID-CUE appear in both projection and decision cases, and SOC-MOR is concentrated in decision-stage cases. This suggests that projection-stage errors are mainly associated with predicting AV behavior, trusting AV capability, and interpreting AV driving style, whereas decision-stage errors are more strongly associated with social and moral evaluation of AVs, perceived exploitability, and intentional priority-taking.

The Factor–Behavior heatmap (Fig. 5b) indicates that predictability, trust, and driving style are frequently co-coded with reduced safety margins and gap or lane-change behavior, while social and moral evaluation is most frequently co-coded with aggression or exploitation. These patterns suggest that human drivers form expectations not only about AV motion, but also about AV agency, capability, cooperativeness, and social status.

Figure 6 shows consistent patterns at the behavior level: Projection-stage cases are concentrated in reduced safety margins and gap or lane-change behavior, while Decision-stage cases are concentrated in aggression or exploitation and right-of-way or yielding behavior.

Together, the three heatmaps in Figs. 5 and 6 reveal a coherent pattern that provides the basis for the two recurring pathways introduced above: projection–stage errors are mainly associated with predicting AV behavior, trusting AV capability, and interpreting AV driving style, producing safety–margin reduction and gap or lane-change behavior; decision-stage errors are more often associated with social and moral evaluation of AVs, producing aggression, exploitation, and right-of-way violations.

5.3 Taxonomy structure

This section synthesizes the case-level attributes defined in Sect. 5.1 and the distributional patterns reported in Sect. 5.2 into the proposed taxonomy structure.

Figure 7 summarizes the proposed spectrum of driver intent in mixed-traffic human–AV interaction by presenting four evidence-informed patterns derived from recurring mechanisms in the coded cases. Their left-to-right ordering is organized along a conceptual continuum from unintentional unsafe adaptation to deliberate strategic exploitation, with the error–violation distinction anchoring the endpoints of this continuum. The spectrum distinguishes relatively unintentional adaptation, such as perceptual contagion or norm transfer from AV platoons and dedicated-lane traffic, or recalibration to the emerging local norm as AV penetration rate and repeated exposure increase, from more deliberate forms of exploitation, such as forcing priority, bullying, brake-checking, tailgating, racing, or blocking. Because the four patterns are grouped by mechanism rather than by stage, perceptual contagion (perception-stage) and penetration/exposure recalibration (projection-stage) are both grouped here as unintentional norm transfer. Between these endpoints are two intermediate mechanisms. The first is prediction error or intent ambiguity, where the driver cannot reliably infer whether the AV will stop, yield, proceed, or accelerate. The second is safety compensation or overtrust, where the driver expects AV sensing, rule compliance, or defensive control to absorb risk. These mechanisms differ in the degree of driver intentionality, but they may produce overlapping observable behaviors, such as short headways, small accepted gaps, delayed deceleration, or unsafe merging.

Figure 8 summarizes the conceptual logic of the taxonomy. AV-specific interaction factors are treated as triggers or modulators that influence different stages of information processing. The resulting observable behaviors are interpreted not as self-explanatory categories, but as outcomes of earlier perception, comprehension, projection, and decision errors that are shaped by AV-specific interaction factors.

The mapping shows that different AV-specific interaction factors are associated with different stages of information processing. Perception-stage cases are mainly linked to AV platoon exposure and AV-dedicated infrastructure, where drivers may copy or normalize short AV/platoon spacing. Comprehension-stage evidence is limited in the current coded set, but it illustrates how AV communication or intent cues may be misinterpreted in multi-agent interactions. Projection-stage cases are mainly associated with AV behavior predictability and intent ambiguity, trust in AV safety and automation capability, AV driving style, and AV recognizability or identity cues. These factors influence how drivers anticipate AV motion, stopping, yielding, acceleration, and collision-avoidance behavior. Decision-stage cases are more strongly associated with AV social status and moral accountability, AV driving style, trust, predictability, and recognizability. These factors influence whether drivers choose to yield, cooperate, exploit, pressure, or deliberately take priority over an AV.

The proposed taxonomy does not replace existing taxonomies of driver errors in conventional traffic. It does not classify human driver errors that would occur independently of AV presence or AV-specific interaction conditions, such as generic distraction, fatigue, impairment, speeding, or rule violations unrelated to interaction with

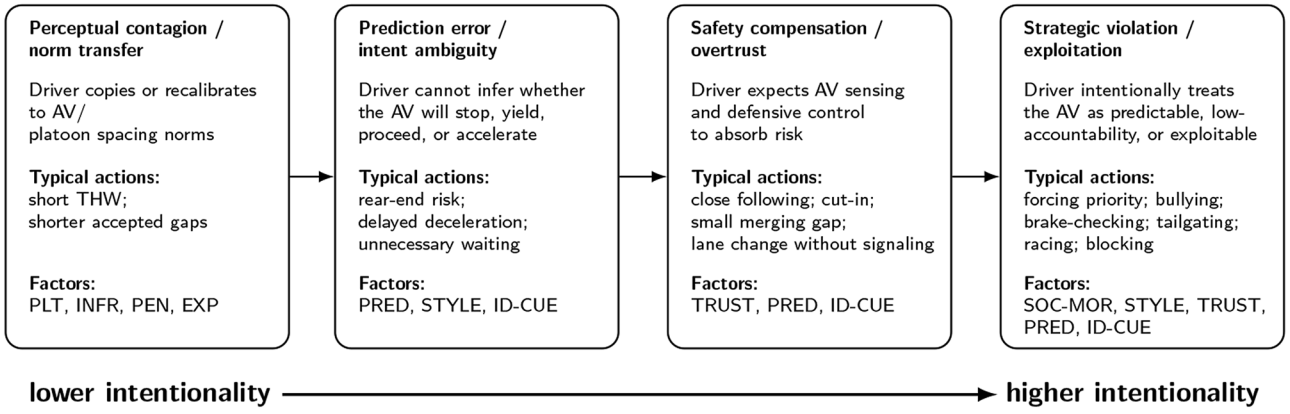


Fig. 7 Driver intent: from unintentional failures to deliberate exploitation of AV behavior

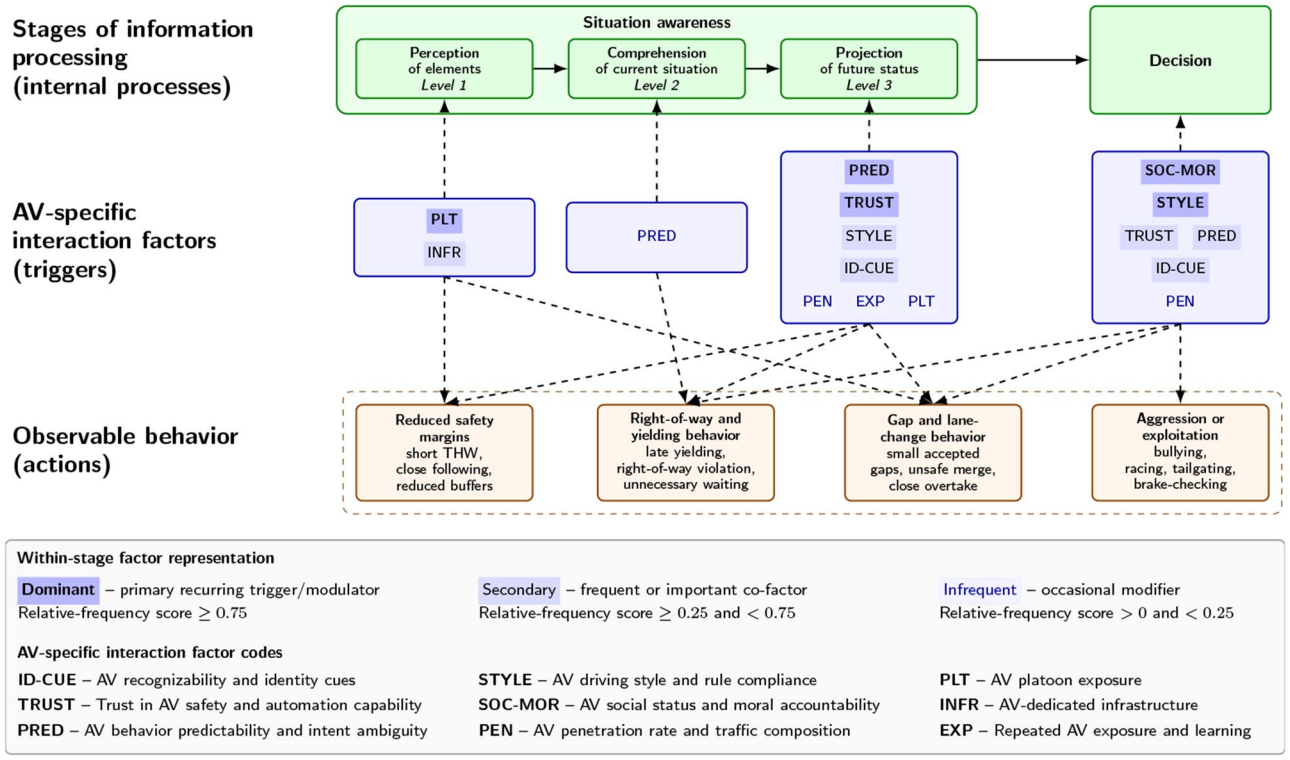


Fig. 8 Taxonomy overview: mapping between AV-specific interaction factors, information-processing stages, and observable behavior. Shading indicates within-stage relative factor representation derived from the coded cases, with thresholds acting as descriptive visualization rules. For each information-processing stage, the count of each factor was divided by the highest factor count observed within that stage, such that the most frequently coded factor received a relative-frequency score of 1.00. Factors with scores of at least 0.75 were classified as *Dominant*, factors with scores from 0.25 to less than 0.75 as *Secondary*, and factors with scores less than 0.25 as *Infrequent*, to provide a descriptive distinction between high-, intermediate-, and low-frequency factors in the coded cases. Dashed arrows indicate the factor–stage and factor–behavior links observed in the synthesis. *Note:* because only one comprehension-stage case was identified, the factor representation shown for this stage should be interpreted illustratively

an AV. Instead, it adds the interaction layer needed for mixed traffic: the AV-specific conditions under which human drivers perceive, interpret, anticipate, and act. This makes the taxonomy useful for organizing empirical evidence, designing mixed-traffic experiments, coding safety-critical events, and identifying AV design choices that may unintentionally shape unsafe behavior among surrounding human drivers.

The taxonomy is therefore intended to support future analysis of mixed-traffic conflicts and accidents, crash data collection, mixed-traffic simulation experiments, and the coding of safety-critical events involving AVs and surrounding human drivers. It also highlights that AV design choices, such as driving style, external communication, recognizability, platooning behavior, rule compliance, and dedicated infrastructure, may shape the error patterns of surrounding human drivers. AV safety should therefore be evaluated not only through the AV's own control performance, but also through its behavioral effect on other road users. More specifically, the attribute structure offers a basis for developing safety performance indicators in mixed traffic, alongside its other uses in conflict analysis, crash-data coding, and simulation design – for example, using AV-specific interaction factors and behavior groups as candidate variables for evidence-based safety assessment.

6 Conclusion

This paper proposes a taxonomy of human driver errors and unsafe adaptations in mixed traffic by linking AV-specific interaction factors to observable unsafe behavior and to the information-processing stages at which errors are inferred to occur. The central argument is that mixed traffic does not necessarily create entirely new human error classes, but changes the conditions under which familiar errors and violations emerge. Human drivers may perceive, interpret, anticipate, and respond to AVs differently from human-driven vehicles, and these differences can reshape the safety implications of otherwise familiar behaviors such as short headways, unsafe merging, delayed yielding, or aggressive driving.

The synthesis of the reviewed evidence suggests that many mixed-traffic errors are concentrated mainly in the projection and decision stages. Projection-stage errors are mainly linked to uncertainty, misjudgment, overtrust, or poor anticipation of AV behavior. These

mechanisms can produce safety-margin reduction, including shorter THW, smaller accepted gaps, closer following, smaller standstill gaps, and reduced overtaking margins. Decision-stage errors are more closely linked to learned expectations, complacency, frustration, social devaluation of AVs, or intentional priority-taking. These mechanisms can produce reduced yielding, aggressive decisions, brake-checking, racing, close cut-ins, and intentional exploitation of conservative AV behavior. The taxonomy highlights two recurring mixed-traffic pathways: projection-stage safety-margin reduction and decision-stage exploitation or social devaluation of AVs. The taxonomy also distinguishes a spectrum of driver intentionality, ranging from relatively unintentional unsafe adaptation or norm transfer to deliberate strategic exploitation of AV behavior.

6.1 Limitations

Several limitations should be noted. First, the taxonomy is based on a synthesis of studies that vary in method, context, and outcome measures. The current evidence base still relies heavily on survey experiments, driving simulators, virtual-reality environments, test-track studies, field experiments, observational datasets, and a limited number of naturalistic or real-world AV interaction studies. These methods make it possible to study behavioral adaptation under controlled conditions, but they also require caution when generalizing to crash causation in natural traffic.

Second, because this was a structured narrative review, the resulting evidence base may be affected by search and selection bias.

Third, the assignment of cases to information-processing stages involves interpretation. The information-processing stage and AV-specific interaction factors cannot always be observed directly. In many cases, they are inferred from the study design, reported behavior, experimental manipulation, driver self-report, traffic context, or the authors' interpretation of the scenario. No formal coding-reliability assessment was conducted. This means that the categorization is partly subjective, particularly for cases near the boundary between projection and decision errors, where the distinction relies on inferred intent rather than observable behavior.

Fourth, the categories were developed and reapplied on the same case set without independent held-out validation, and the identified AV-specific interaction factors should therefore not be considered exhaustive. The factor set is evidence-dependent, and additional interaction factors may emerge as the mixed-traffic evidence base expands.

Fifth, taken together, the limitations above regarding search and selection bias, the absence of a formal coding-reliability assessment, and the absence of independent held-out validation mean that the results reported here should not be assumed to be exactly reproducible. An independent team following the same eligibility criteria and coding definitions (Sect. 2 and Sect. 5.1) may identify a different set of cases or assign some cases differently, particularly near category boundaries. The overall taxonomy structure and dominant patterns therefore require independent reapplication and validation beyond the present evidence set.

Sixth, the distribution of cases, presented in Fig. 2, should not be interpreted as the true prevalence of error types in real mixed traffic. The cases included in this review and their counts reflect the research emphasis of the available literature and the coding approach used in this synthesis. Some factors may appear more frequently because they have been studied more often, while others may be underrepresented because they are difficult to observe, isolate, or experimentally manipulate.

Finally, the taxonomy focuses on human drivers of manually driven vehicles interacting with AVs as surrounding road users. It does not cover pedestrians, cyclists, AV passengers, remote operators, or drivers supervising partially automated systems. These traffic participants may experience different forms of behavioral adaptation. The taxonomy may provide a starting point for broader mixed-traffic classification, but it should not be assumed to generalize directly to all road-user groups.

6.2 Future work

Future research should focus on validating and refining the proposed taxonomy using empirical data. Naturalistic-driving studies and real-world mixed-traffic datasets provide opportunities to examine behavioral adaptation under realistic conditions, especially when vehicle trajectories, video, contextual information, and driver behavior can be linked. Such data would help test whether the projection-stage safety-margin reduction and decision-stage exploitation patterns identified here are also visible in real safety-critical events.

The taxonomy can also inform AV design and policy. If certain AV behaviors trigger unsafe human adaptation, then AV safety assessment should include not only the AV's own compliance and collision-avoidance performance, but also its behavioral effect on surrounding drivers. Mixed-traffic safety should therefore be evaluated as an interaction outcome: the result of AV behavior, human expectations, traffic composition, and infrastructure. In this sense, the taxonomy provides a basis for evidence-based mixed-traffic safety research and for designing AVs that are not only technically safe but also behaviorally compatible with human road users.

Supplementary information

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Supplementary Material 1

Author contributions

O.M. wrote the main manuscript text, prepared figures, and tables. All authors reviewed the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

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